

RAG and Agentic AI — Practice Notes

These notes are created for practicing a Retrieval-Augmented Generation (RAG) chatbot. The chatbot should answer questions using the information contained in this document.

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks that normally require human-like capabilities. These tasks can include understanding language, recognizing patterns, making predictions, and solving problems.

2. What is a Large Language Model?

A Large Language Model (LLM) is a machine learning model trained on large amounts of text. It can generate text, answer questions, summarize information, transform text, and perform many language-related tasks. An LLM does not automatically have access to a user's private documents.

3. What is RAG?

Retrieval-Augmented Generation, commonly called RAG, is an architecture that combines information retrieval with text generation. Instead of relying only on the language model's internal knowledge, RAG retrieves relevant information from an external knowledge source and provides that information to the LLM as context.

4. RAG Pipeline

A typical RAG pipeline has two major stages. During indexing, documents are loaded, split into smaller chunks, converted into embeddings, and stored in a vector database. During retrieval, a user question is converted into an embedding, similar chunks are retrieved, and those chunks are provided to the language model along with the question.

5. Document Loaders

A document loader imports information from a source into a format that an application can process. For example, a PDF loader can read text from PDF pages. LangChain provides loaders for different types of sources such as PDFs, text files, web pages, CSV files, and other data sources.

6. Chunking

Chunking means dividing a large document into smaller pieces. Smaller chunks make it easier to find specific information during retrieval. Chunk size and chunk overlap are important settings. Overlap allows neighboring chunks to share some text so that useful context is not lost at chunk boundaries.

7. Embeddings

An embedding is a numerical representation of text. Texts with similar meanings tend to have similar vector representations. Embeddings allow a vector database to perform semantic similarity searches.

8. Vector Database

A vector database stores vector representations and allows applications to search for vectors that are similar to a query vector. Chroma is one example of a vector database that can be used in RAG systems.

9. Retriever

A retriever is responsible for finding relevant documents or document chunks for a user's query. It does not normally generate the final answer. Its job is to retrieve useful context that can be passed to another component such as an LLM.

10. Retrieval-Augmented Answering

When a user asks a question, the application retrieves relevant chunks from the knowledge base. The question and retrieved context are then placed into a prompt. The LLM uses that context to generate an answer. A well-designed RAG system should prefer the retrieved source information when answering questions about the connected documents.

11. Example Questions

Useful questions for testing a RAG chatbot include: What is RAG? What does a document loader do? Why is chunking necessary? What is an embedding? What is the purpose of a vector database? What does a retriever do? What are the two major stages of a typical RAG pipeline?

12. Key Terms

AI: Artificial Intelligence. LLM: Large Language Model. RAG: Retrieval-Augmented Generation. Embedding: numerical representation of text. Chunk: a smaller piece of a document. Retriever: component that finds relevant information. Vector database: database designed to store and search vector representations.